

DHCP: Dynamic Host Configuration Protocol

IP addresses: how to get one?

That's actually **two** questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)?

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: Dynamic Host Configuration Protocol: dynamically get address from as server
 - “plug-and-play”

IP addresses: how to get one?

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 197 . 168 . 10 . 17

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 197 . 168 . 10 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 197 . 168 . 10 . 8

Alternate DNS server: 8 . 8 . 8 . 8

☐ Validate settings upon exit

Advanced...

OK Cancel

Internet Protocol Version 4 (TCP/IPv4) Properties

General Alternate Configuration

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address: . . .

Subnet mask: . . .

Default gateway: . . .

☒ Obtain DNS server address automatically

☐ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

IP Cont..

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix  . : iitbhilai.ac.in
Description . . . . . : Qualcomm QCA9377 802.11ac Wireless Adapter
Physical Address. . . . . : B0-68-E6-82-D9-8D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::80d1:147e:1fc0:c043%9(Preferred)
IPv4 Address. . . . . : 10.3.54.107(Preferred)
Subnet Mask . . . . . : 255.255.192.0
Lease Obtained. . . . . : 29 September 2020 09:39:45
Lease Expires . . . . . : 29 September 2020 18:28:38
Default Gateway . . . . . : 10.3.0.1
DHCP Server . . . . . : 10.1.72.7
DHCPv6 IAID . . . . . : 162556134
DHCPv6 Client DUID. . . . . : 00-01-00-01-24-F3-49-C3-D8-D0-90-5B-50-B5
DNS Servers . . . . . : 192.168.10.87
                        192.168.10.72
```

DHCP: Dynamic Host Configuration Protocol

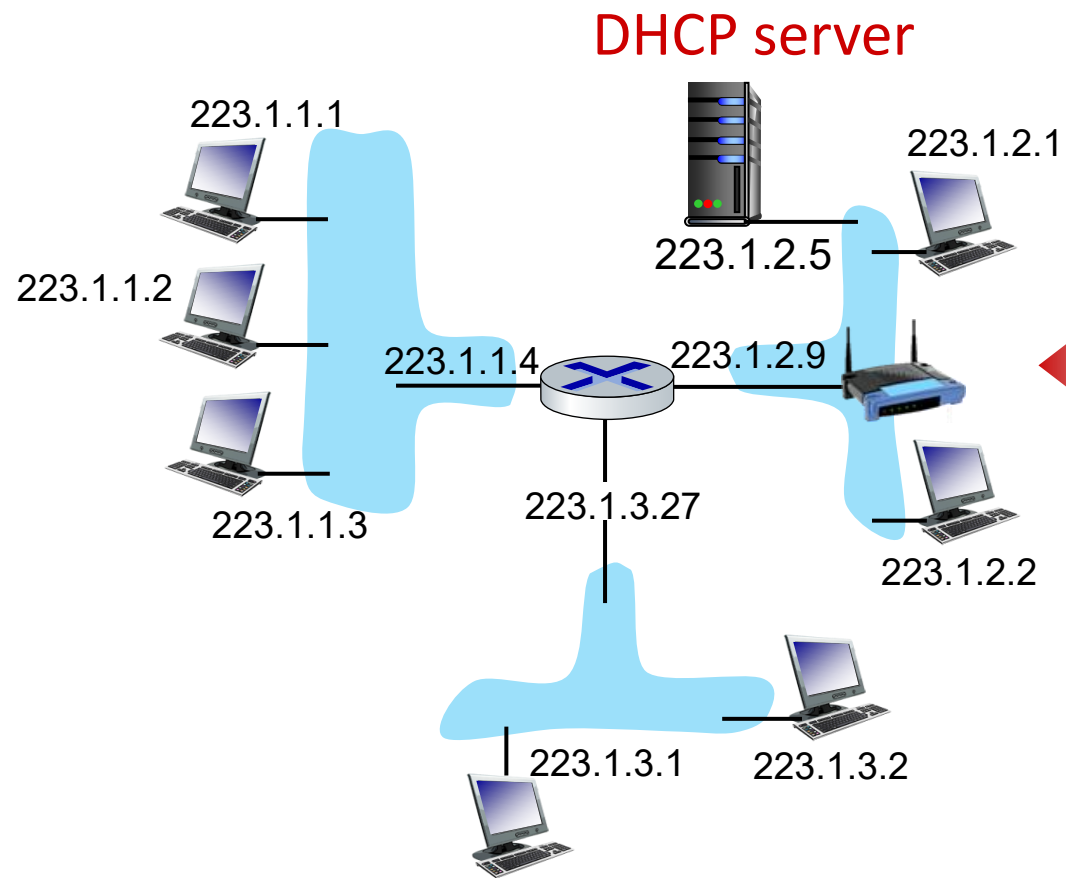
goal: host *dynamically* obtains IP address from network server when it “joins” network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg

DHCP client-server scenario



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



arriving **DHCP client** needs address in this network

DHCP client-server scenario

DHCP server: 223.1.2.5



DHCP discover

Broadcast: is there a
DHCP server out there?

Arriving client



DHCP offer

Broadcast: I'm a DHCP
server! Here's an IP
address you can use

DHCP request

Broadcast: OK. I would
like to use this IP address!

DHCP ACK

Broadcast: OK. You've
got that IP address!

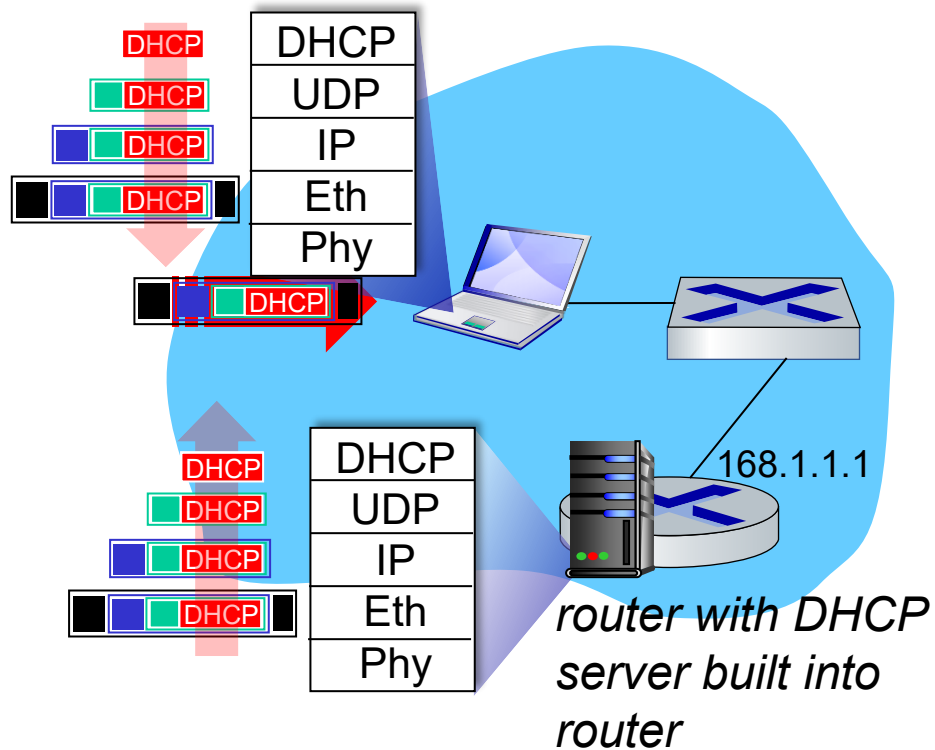
The two steps above can
be skipped "if a client
remembers and wishes to
reuse a previously
allocated network address"
[RFC 2131]

DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

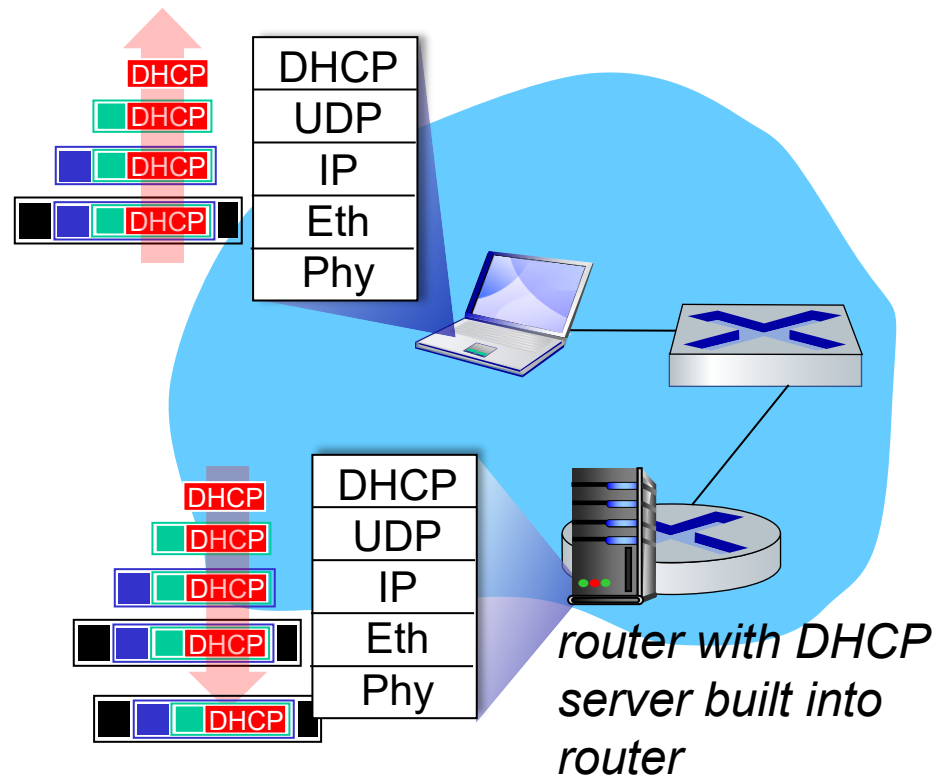
- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

DHCP: example



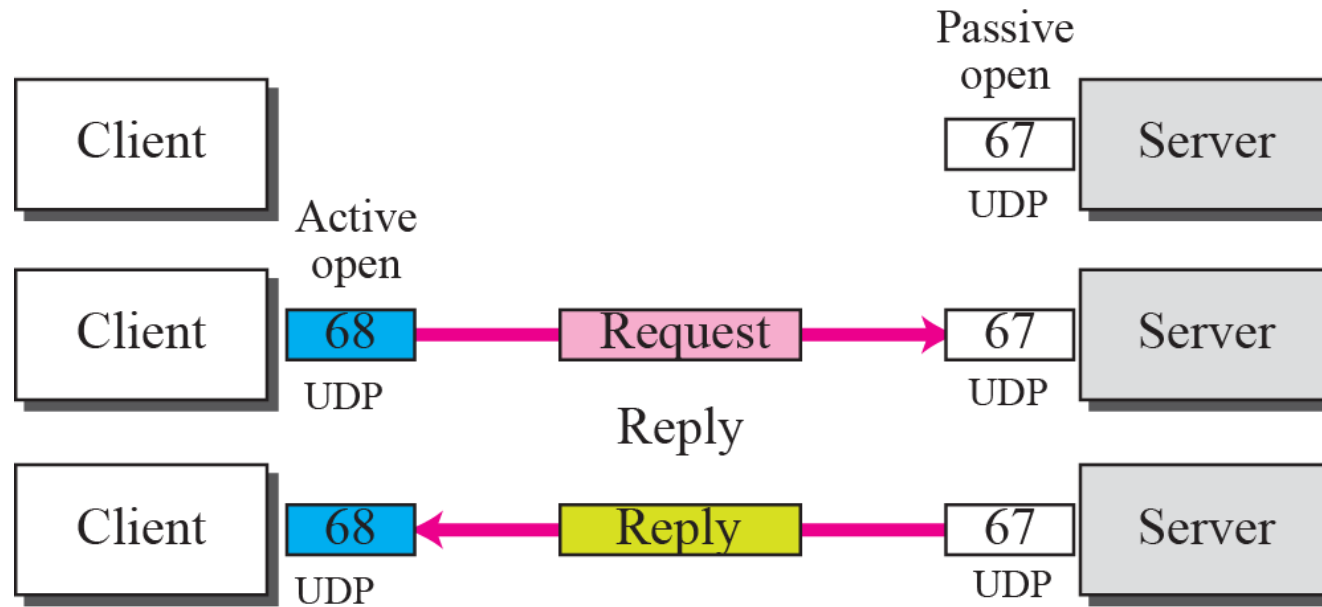
- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP

DHCP: example



- DHCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

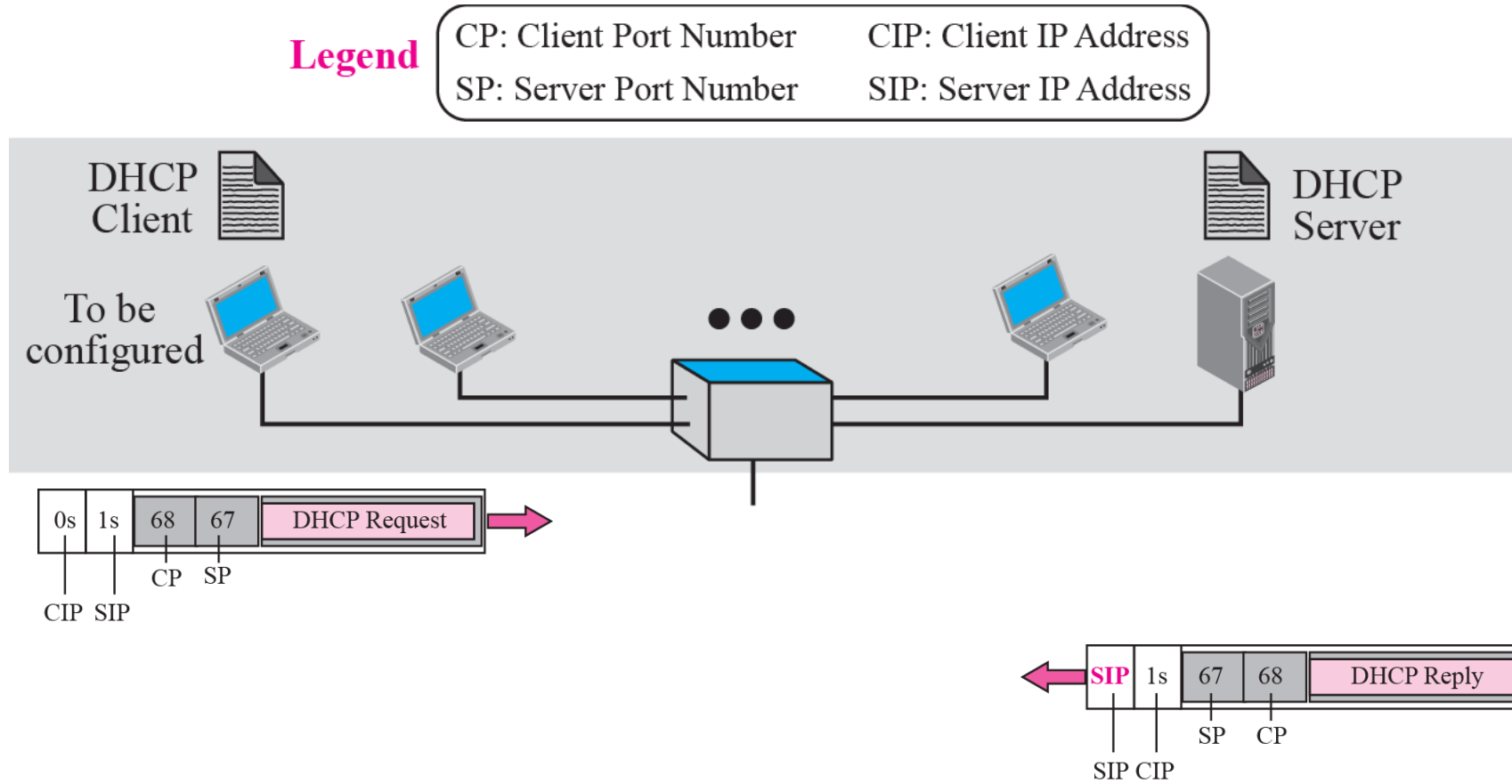
Use of UDP ports



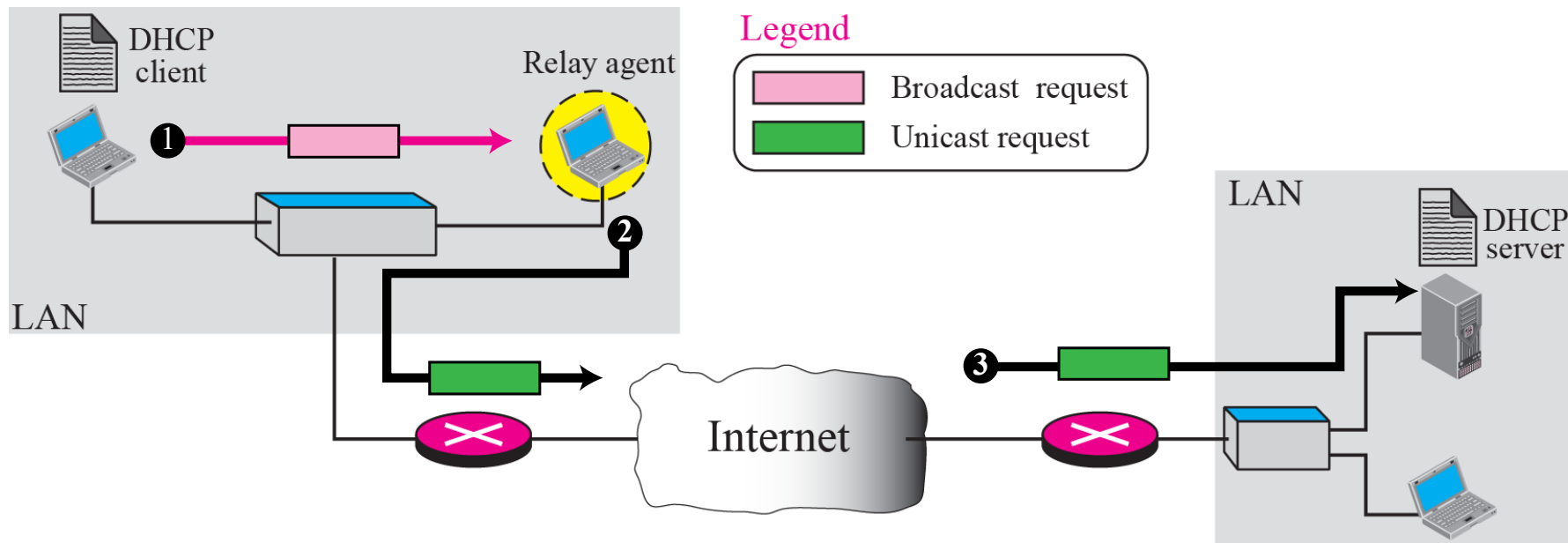
DHCP Operation

The DHCP client and server can either be on the same network or on different networks. Let us discuss each situation separately.

Client and server on the same network

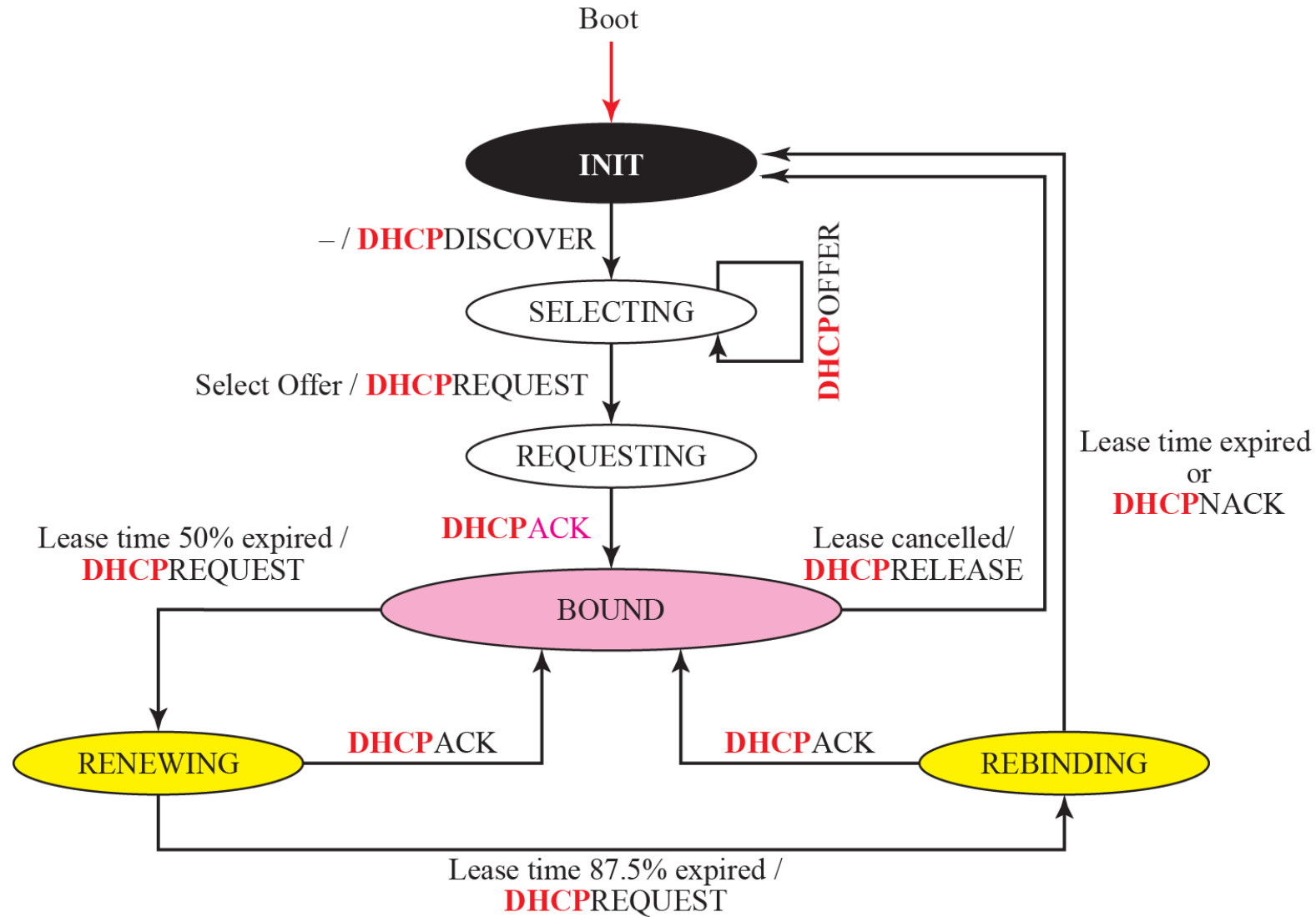


Client and server on two different networks

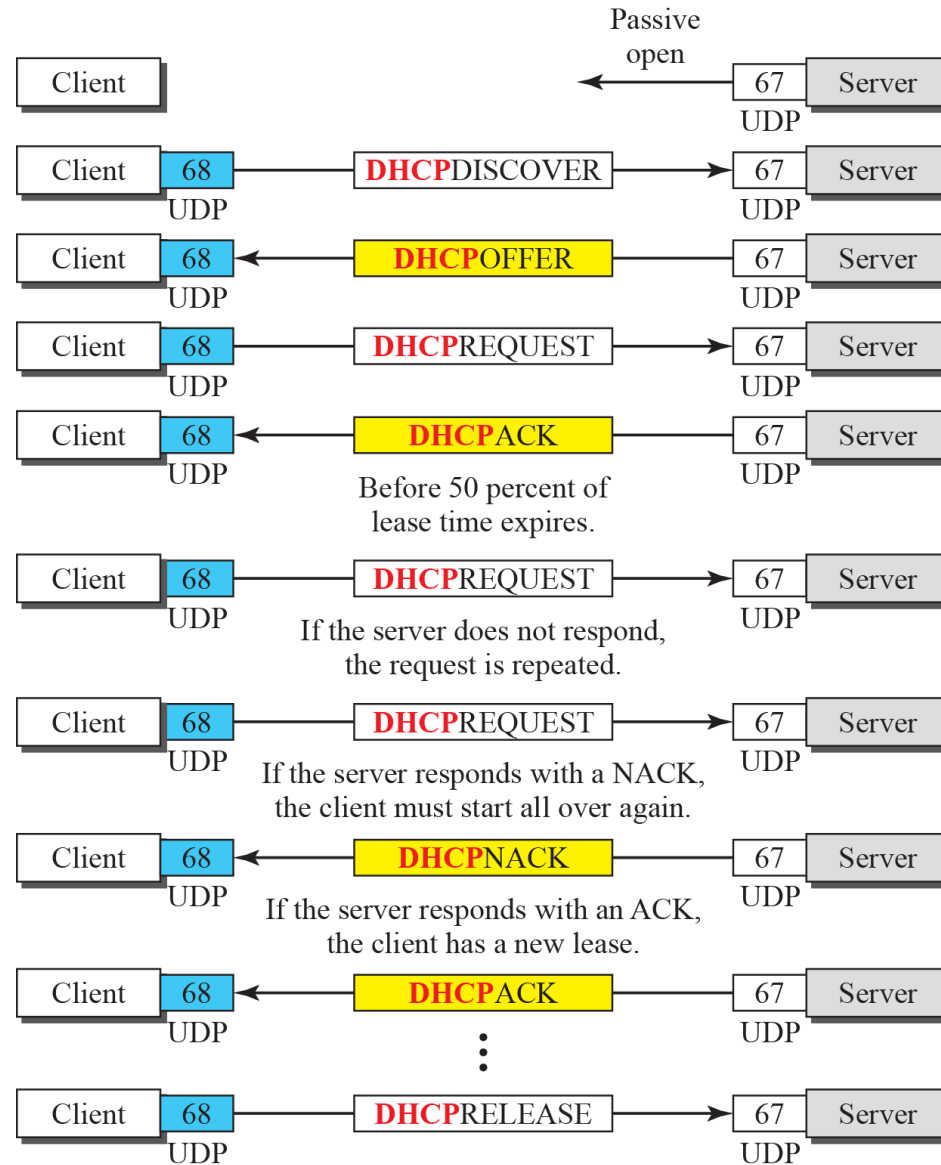


Usually, Routers/layer 3 switches (default gateway of the subnet) act as relay agent

DHCP client transition diagram



Exchanging messages



IP addresses: how to get one?

Q: how does *network* get subnet part of IP address?

A: gets allocated portion of its provider ISP's address space

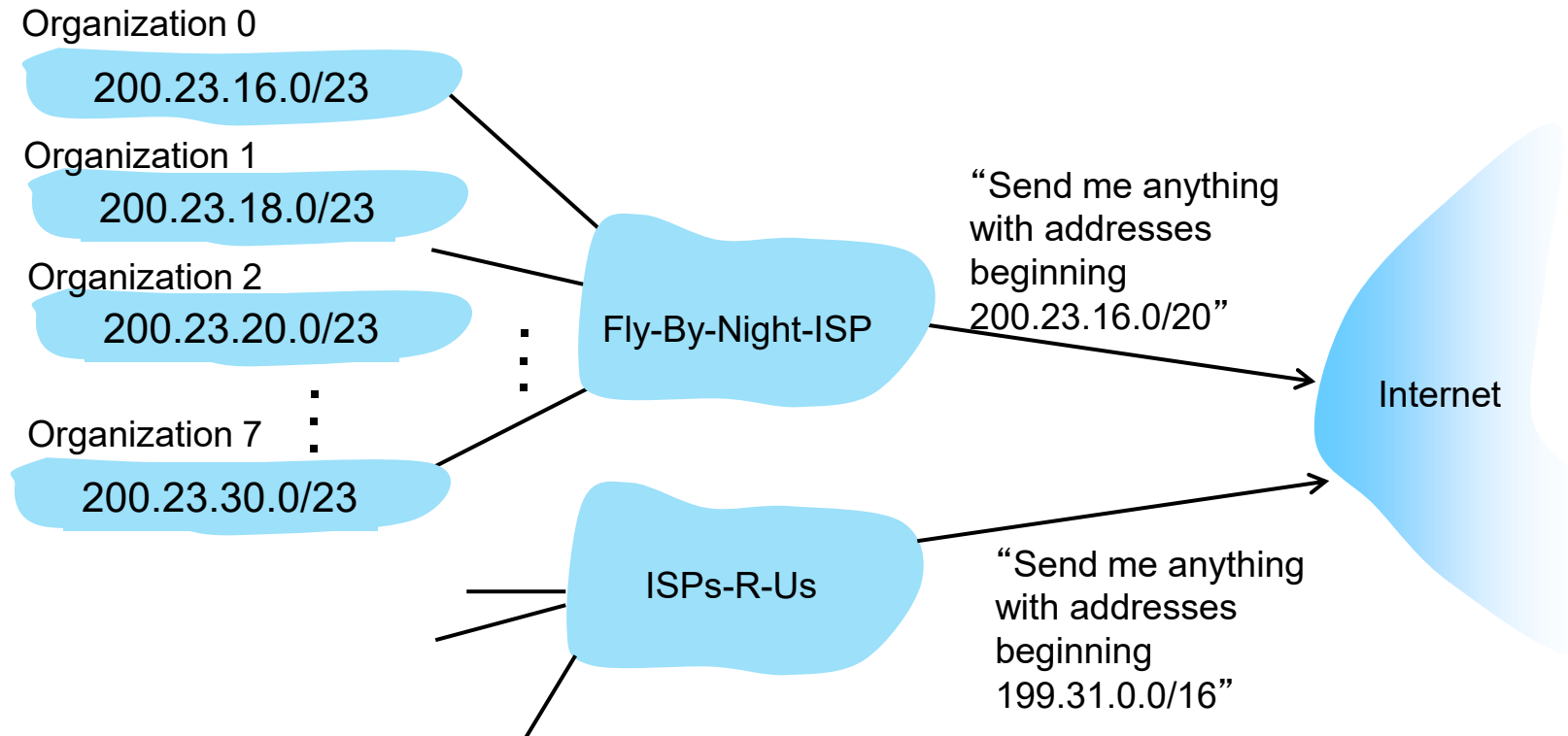
ISP's block 11001000 00010111 00010000 00000000 200.23.16.0/20

ISP can then allocate out its address space in 8 blocks:

Organization 0	<u>11001000 00010111 00010000</u>	00000000	200.23.16.0/23
Organization 1	<u>11001000 00010111 00010010</u>	00000000	200.23.18.0/23
Organization 2	<u>11001000 00010111 00010100</u>	00000000	200.23.20.0/23
...			
Organization 7	<u>11001000 00010111 00011110</u>	00000000	200.23.30.0/23

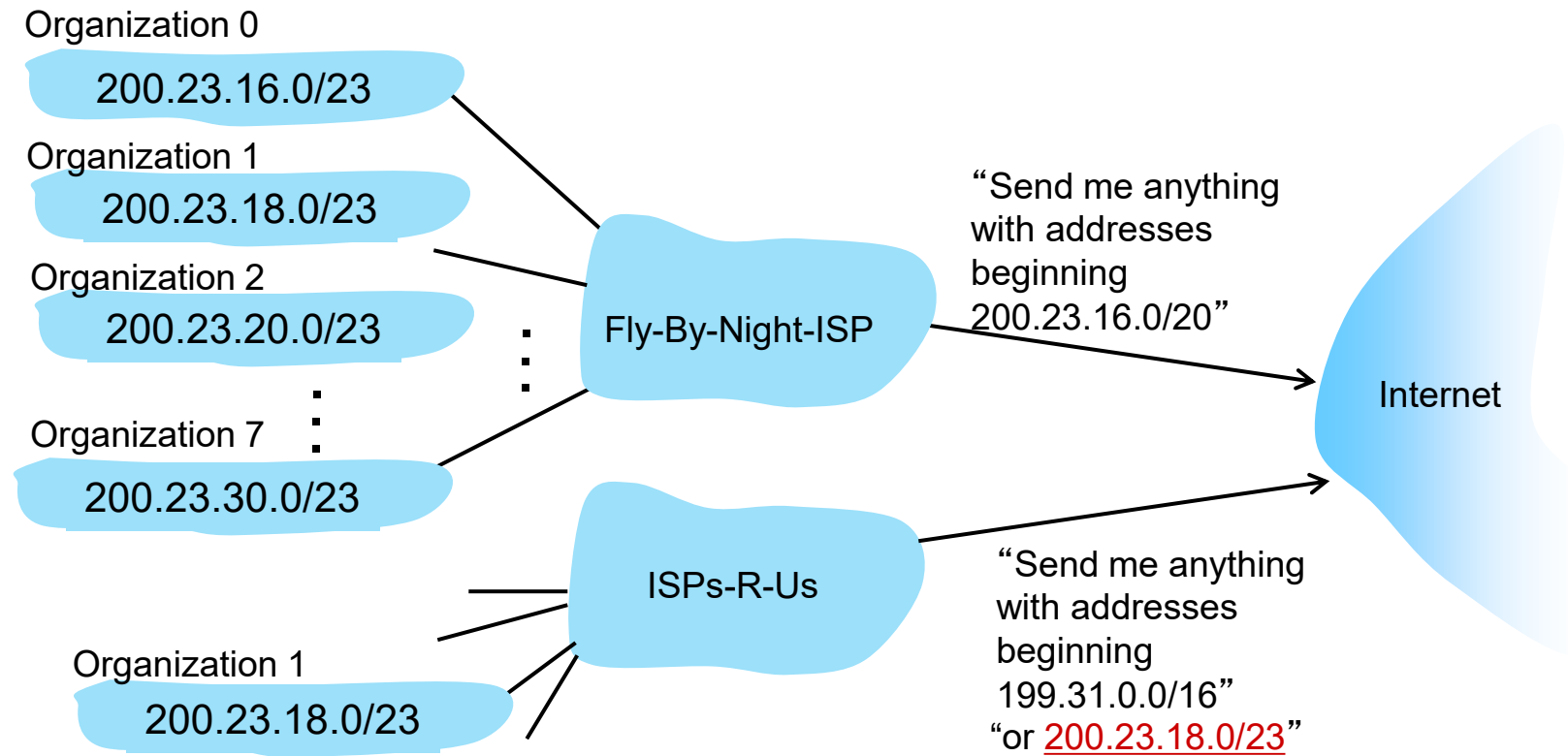
Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:



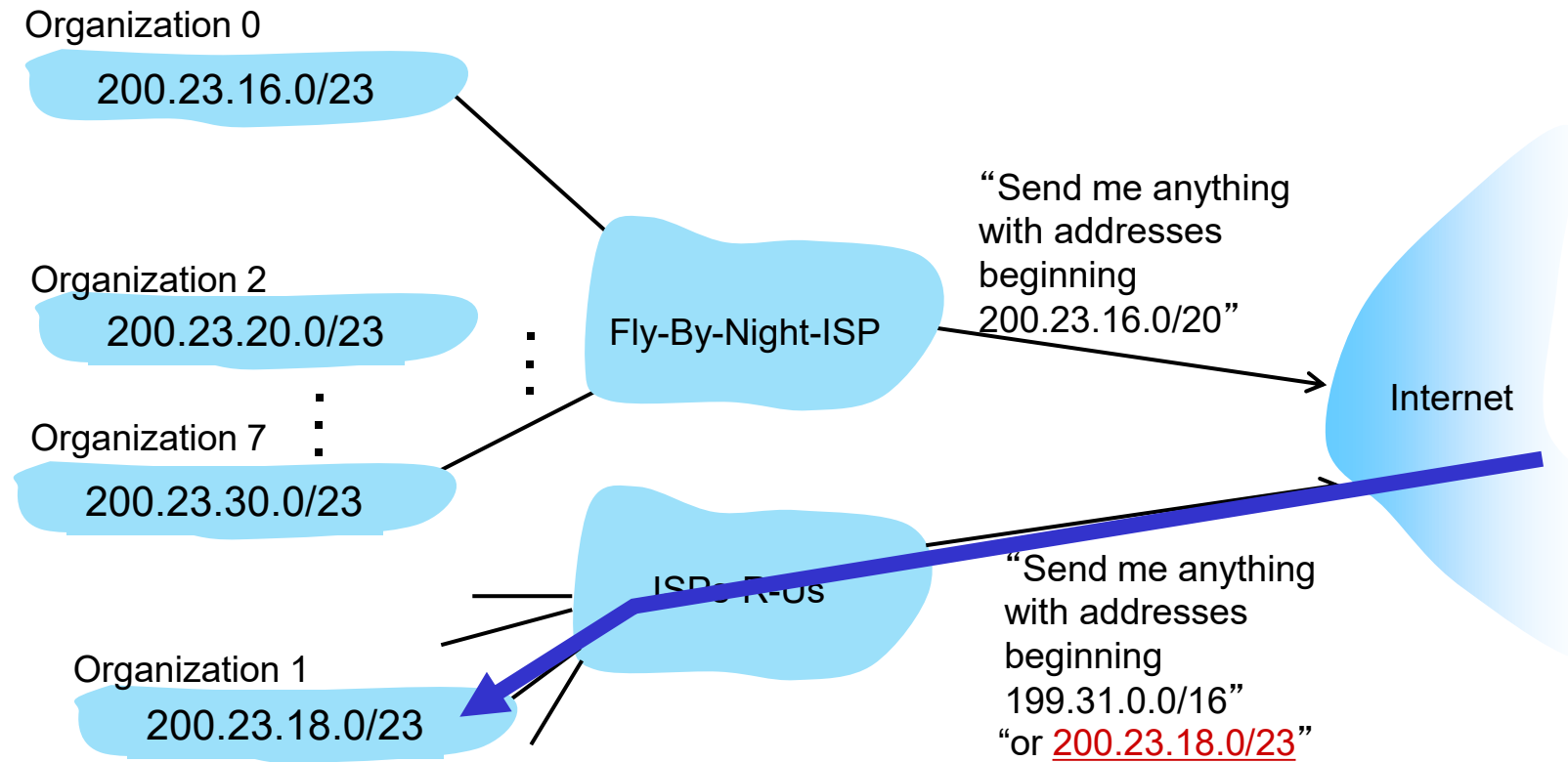
Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



IP addressing: last words ...

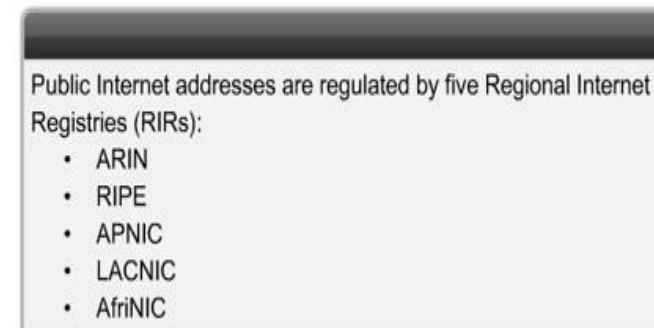
Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers
<http://www.icann.org/>

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space



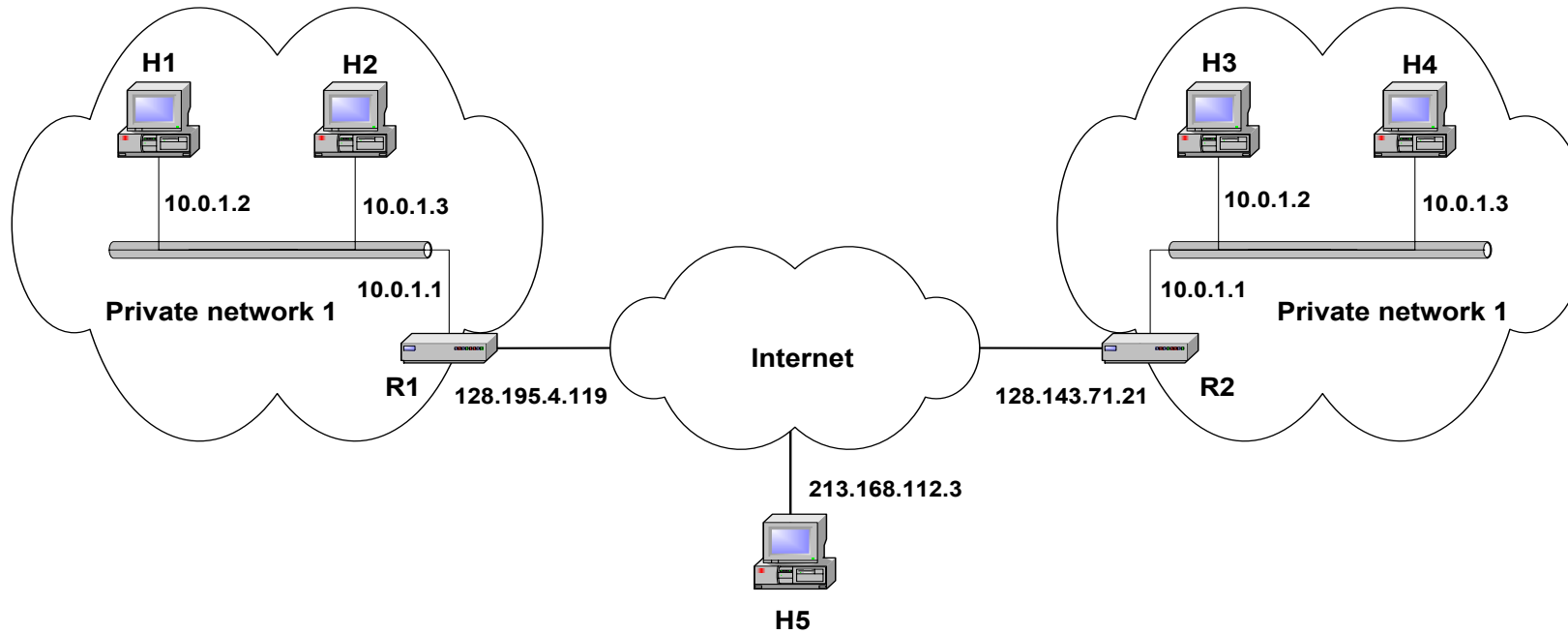
NAT: Network Address Translation

Private Network

- *Private IP* network is an IP network that is not directly connected to the Internet
- IP addresses in a private network can be assigned arbitrarily.
 - Not registered and not guaranteed to be globally unique
- Generally, private networks use addresses from the following experimental address ranges (**Free and *non-routable* addresses**): **[RFC 1918]**
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

Private Addresses



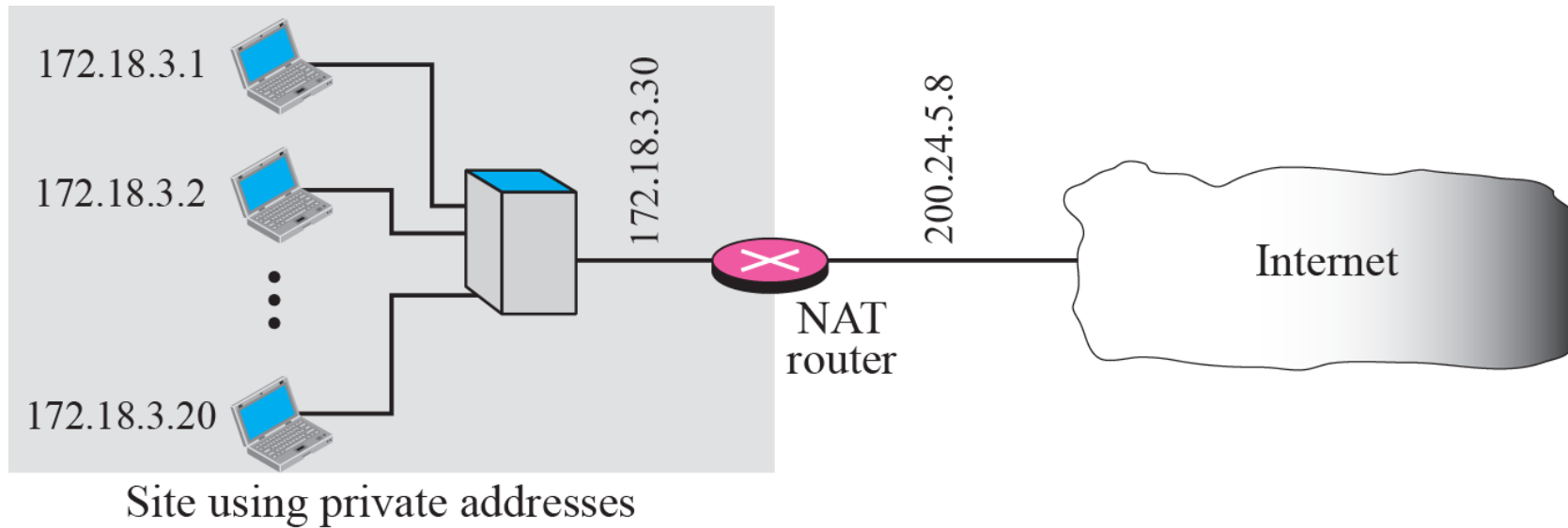
Network Address Translation (NAT)

- RFC 1631
- A short term solution to the problem of the depletion of IP addresses
 - Long term solution is IP v6
 - CIDR (Classless Inter Domain Routing) is a possible short term solution
 - NAT is another
- NAT is a way to conserve IP addresses
 - Can be used to hide a number of hosts behind a single IP address
 - Uses private addresses:
 - 10.0.0.0-10.255.255.255,
 - 172.16.0.0-172.32.255.255 or
 - 192.168.0.0-192.168.255.255

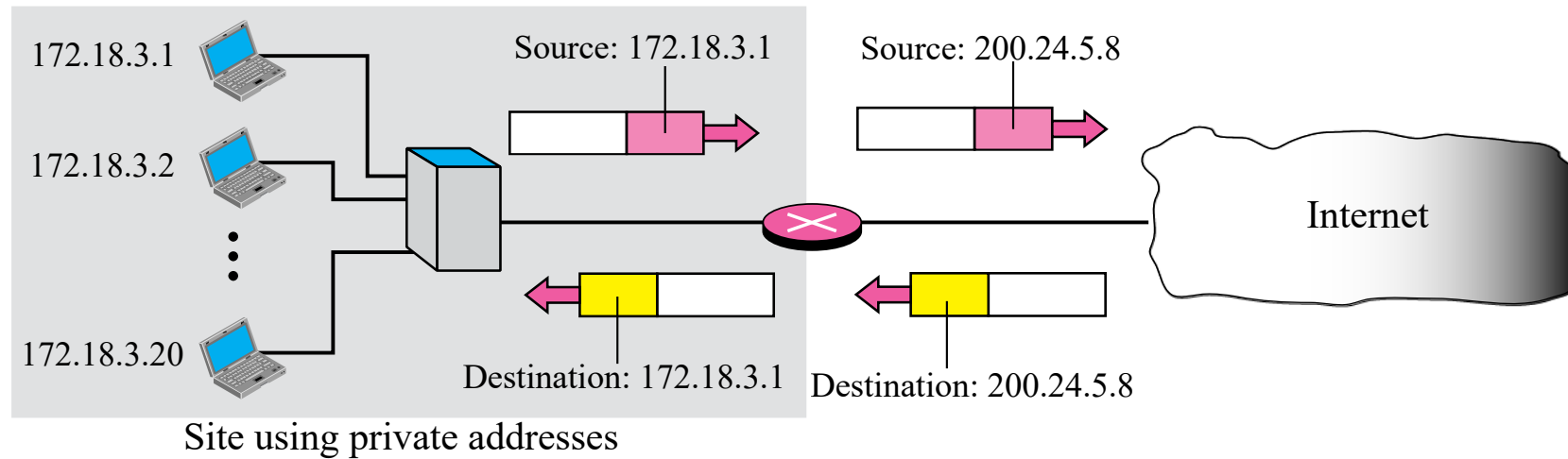
Network Address Translation (NAT)

- NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network
- NAT is a method that enables hosts on private networks to communicate with hosts on the Internet
- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.
- NAT translates non-routable, private, internal addresses into routable, public addresses.
- NAT has an added benefit of adding a degree of privacy and security to a network because it hides internal IP addresses from outside networks.

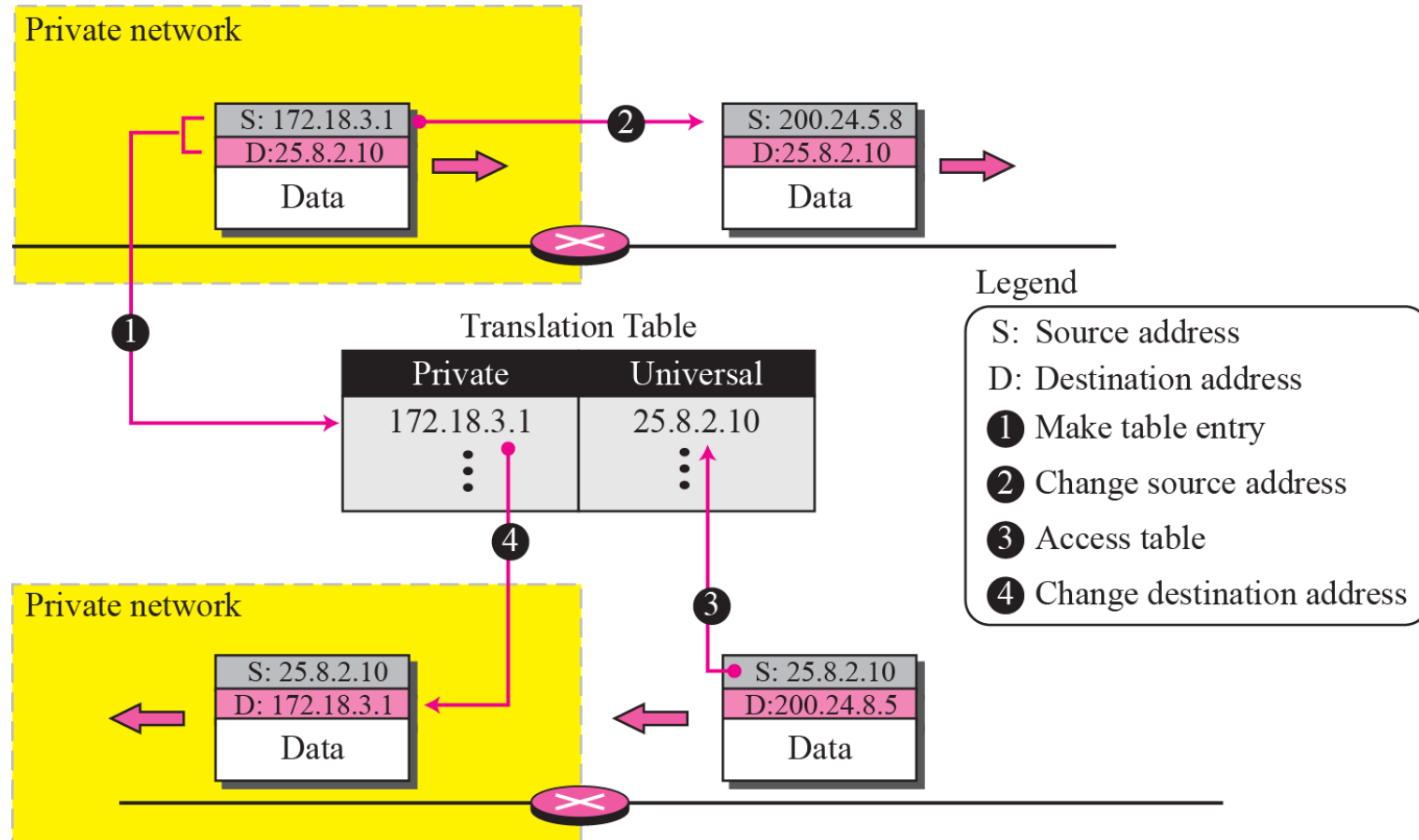
Network Address Translation (NAT)



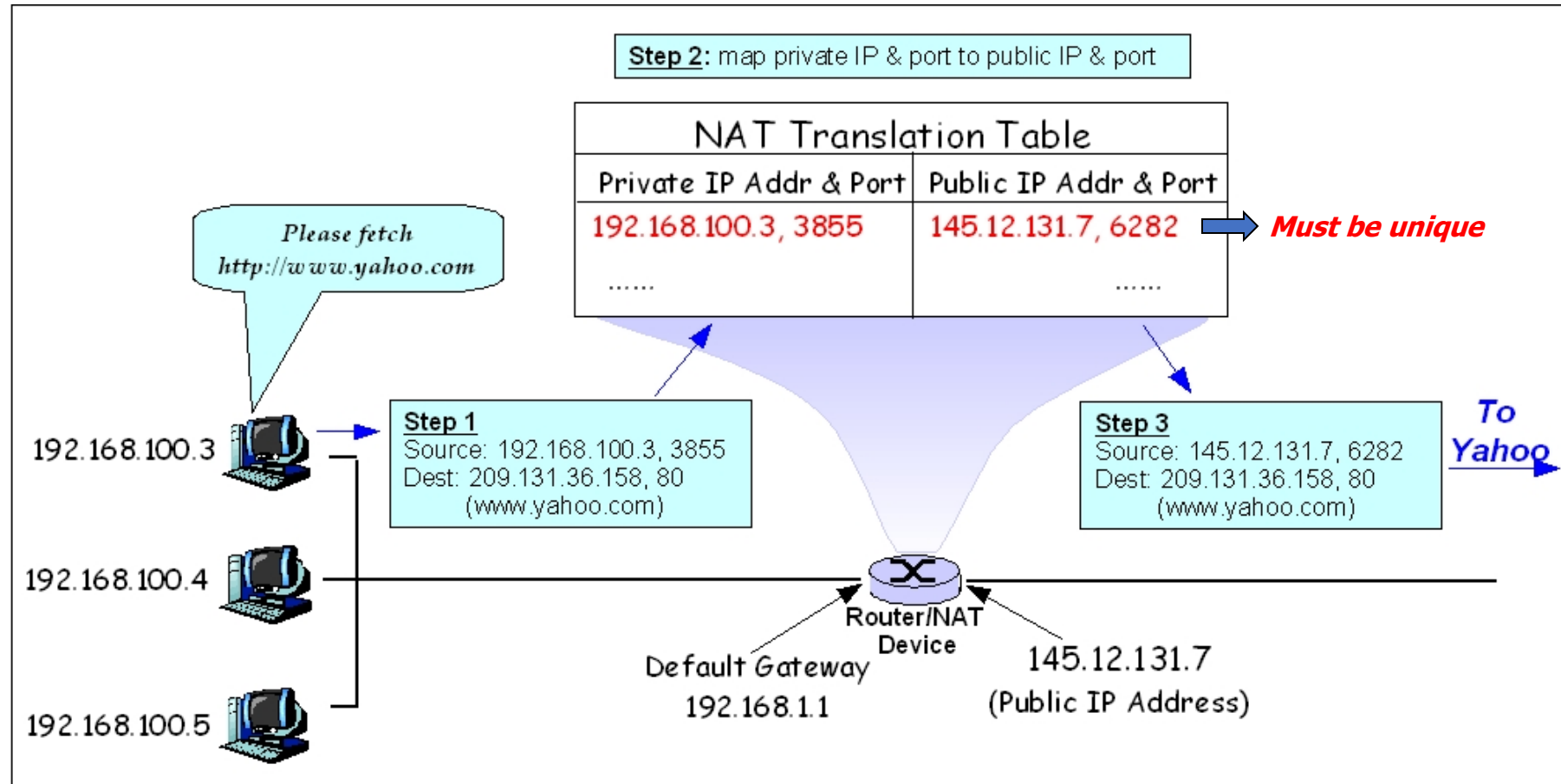
Address Resolution



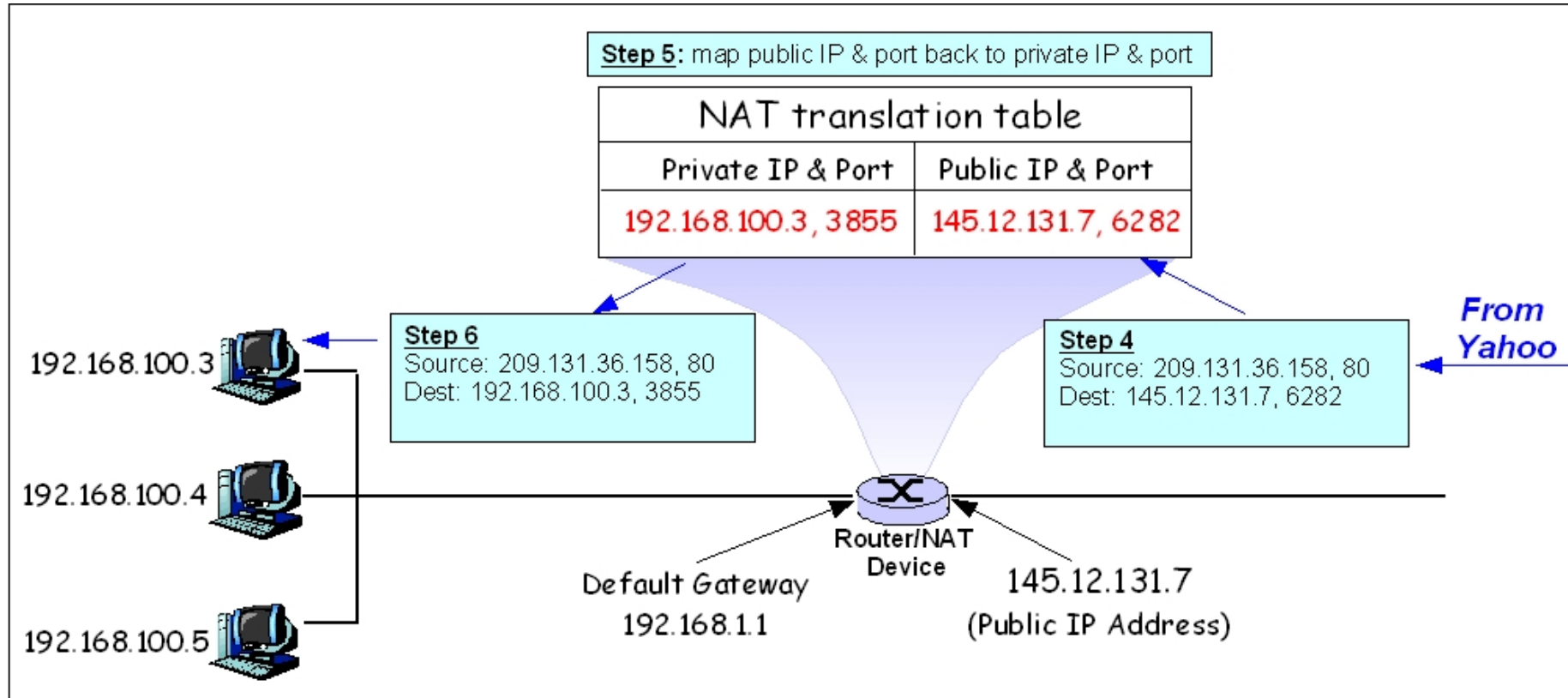
Translation



NAT Table with IP address & Port



NAT Table with IP address & Port



Concerns about NAT

- **Performance:**
 - Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
 - Modifying port number requires that NAT boxes recalculate TCP checksum
- **End-to-end connectivity:**
 - NAT destroys universal end-to-end reachability of hosts on the Internet.
 - A host in the public Internet often cannot initiate communication to a host in a private network.
 - The problem is worse, when two hosts that are in a private network need to communicate with each other.